



1. Description

1.1. Project

Project Name	STM32CubeMX
Board Name	custom
Generated with:	STM32CubeMX 6.17.0
Date	07/08/2026

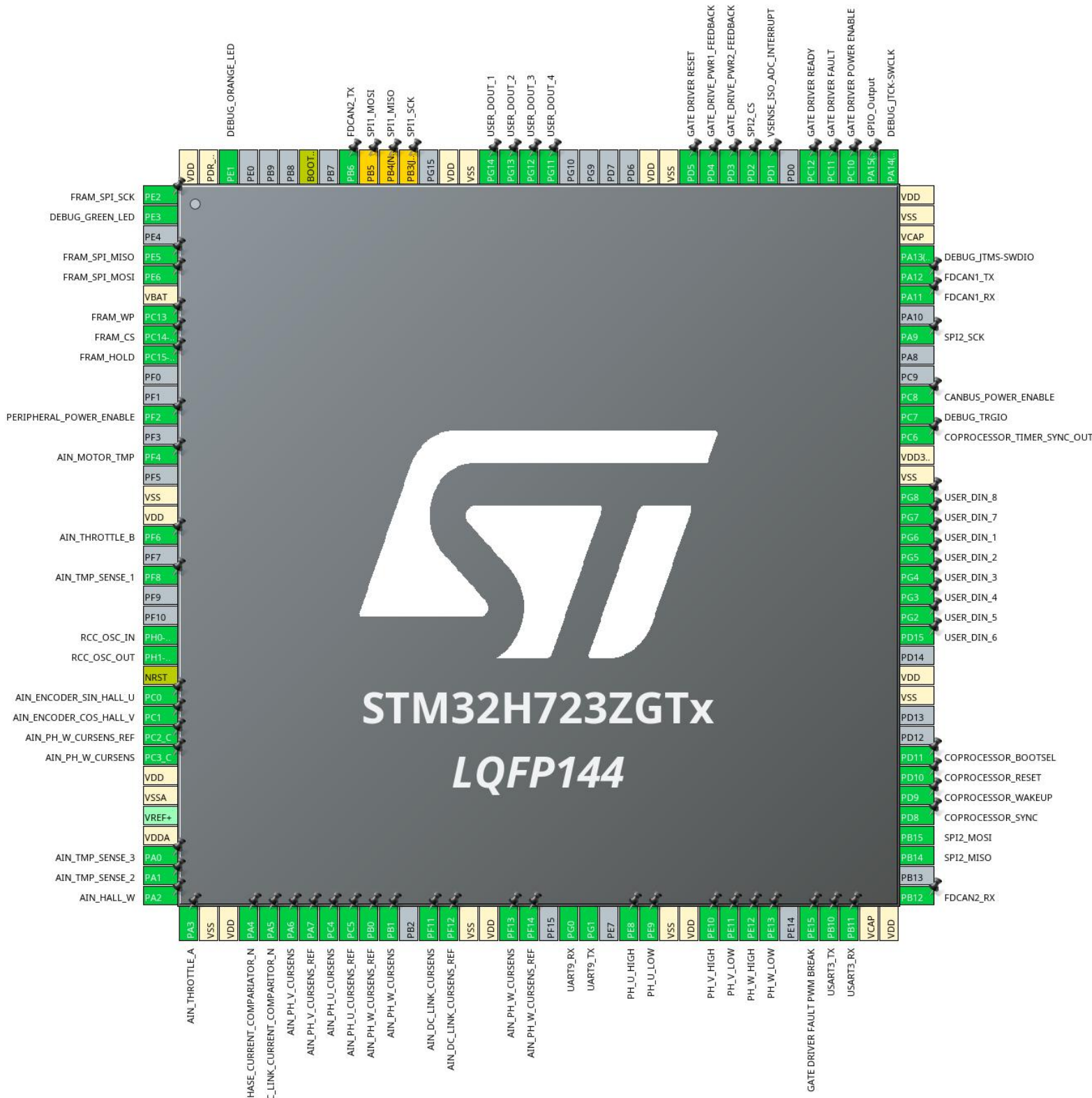
1.2. MCU

MCU Series	STM32H7
MCU Line	STM32H723/733
MCU name	STM32H723ZGTx
MCU Package	LQFP144
MCU Pin number	144

1.3. Core(s) information

Core(s)	Arm Cortex-M7
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2. Pinout Configuration



3. Pins Configuration

Pin Number LQFP144	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	PE2	I/O	SPI4_SCK	FRAM_SPI_SCK
2	PE3 *	I/O	GPIO_Output	DEBUG_GREEN_LED
4	PE5	I/O	SPI4_MISO	FRAM_SPI_MISO
5	PE6	I/O	SPI4_MOSI	FRAM_SPI_MOSI
6	VBAT	Power		
7	PC13 *	I/O	GPIO_Output	FRAM_WP
8	PC14-OSC32_IN *	I/O	GPIO_Output	FRAM_CS
9	PC15-OSC32_OUT *	I/O	GPIO_Output	FRAM_HOLD
12	PF2 *	I/O	GPIO_Output	PERIPHERAL_POWER_EN ABLE
14	PF4	I/O	ADC3_INP9	AIN_MOTOR_TMP
16	VSS	Power		
17	VDD	Power		
18	PF6	I/O	ADC3_INP8	AIN_THROTTLE_B
20	PF8	I/O	ADC3_INP7	AIN_TMP_SENSE_1
23	PH0-OSC_IN	I/O	RCC_OSC_IN	
24	PH1-OSC_OUT	I/O	RCC_OSC_OUT	
25	NRST	Reset		
26	PC0	I/O	ADC2_INP10	AIN_ENCODER_SIN_HALL _U
27	PC1	I/O	ADC2_INP11	AIN_ENCODER_COS_HAL L_V
28	PC2_C	I/O	ADC3_INN1, ADC3_INP0	AIN_PH_W_CURSENS_RE F
29	PC3_C	I/O	ADC3_INP1	AIN_PH_W_CURSENS
30	VDD	Power		
31	VSSA	Power		
33	VDDA	Power		
34	PA0	I/O	ADC1_INP16	AIN_TMP_SENSE_3
35	PA1	I/O	ADC1_INP17	AIN_TMP_SENSE_2
36	PA2	I/O	ADC2_INP14	AIN_HALL_W
37	PA3	I/O	ADC2_INP15	AIN_THROTTLE_A
38	VSS	Power		
39	VDD	Power		
40	PA4	I/O	DAC1_OUT1	PHASE_CURRENT_COMP ARIATOR_N

Pin Number LQFP144	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
41	PA5	I/O	DAC1_OUT2	DC_LINK_CURRENT_COM PARITOR_N
42	PA6	I/O	ADC1_INP3, ADC2_INP3	AIN_PH_V_CURSENS
43	PA7	I/O	ADC1_INN3, ADC2_INN3, ADC1_INP7, ADC2_INP7	AIN_PH_V_CURSENS_RE F
44	PC4	I/O	ADC1_INP4	AIN_PH_U_CURSENS
45	PC5	I/O	ADC1_INN4, ADC1_INP8, ADC2_INP8	AIN_PH_U_CURSENS_RE F
46	PB0	I/O	ADC1_INN5, ADC1_INP9, ADC2_INP9	AIN_PH_W_CURSENS_RE F
47	PB1	I/O	ADC1_INP5	AIN_PH_W_CURSENS
49	PF11	I/O	ADC1_INP2	AIN_DC_LINK_CURSENS
50	PF12	I/O	ADC1_INN2, ADC1_INP6	AIN_DC_LINK_CURSENS_ REF
51	VSS	Power		
52	VDD	Power		
53	PF13	I/O	ADC2_INP2	AIN_PH_W_CURSENS
54	PF14	I/O	ADC2_INN2, ADC2_INP6	AIN_PH_W_CURSENS_RE F
56	PG0	I/O	UART9_RX	
57	PG1	I/O	UART9_TX	
59	PE8	I/O	TIM1_CH1N	PH_U_HIGH
60	PE9	I/O	TIM1_CH1	PH_U_LOW
61	VSS	Power		
62	VDD	Power		
63	PE10	I/O	TIM1_CH2N	PH_V_HIGH
64	PE11	I/O	TIM1_CH2	PH_V_LOW
65	PE12	I/O	TIM1_CH3N	PH_W_HIGH
66	PE13	I/O	TIM1_CH3	PH_W_LOW
68	PE15	I/O	TIM1_BKIN	GATE DRIVER FAULT PWM BREAK
69	PB10	I/O	USART3_TX	
70	PB11	I/O	USART3_RX	
71	VCAP	Power		
72	VDD	Power		
73	PB12	I/O	FDCAN2_RX	
75	PB14	I/O	SPI2_MISO	
76	PB15	I/O	SPI2_MOSI	
77	PD8 *	I/O	GPIO_Output	COPROCESSOR_SYNC
78	PD9 *	I/O	GPIO_Output	COPROCESSOR_WAKEUP
79	PD10 *	I/O	GPIO_Output	COPROCESSOR_RESET

Pin Number LQFP144	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
80	PD11 *	I/O	GPIO_Output	COPROCESSOR_BOOTSEL
83	VSS	Power		
84	VDD	Power		
86	PD15 *	I/O	GPIO_Input	USER_DIN_6
87	PG2 *	I/O	GPIO_Input	USER_DIN_5
88	PG3 *	I/O	GPIO_Input	USER_DIN_4
89	PG4 *	I/O	GPIO_Input	USER_DIN_3
90	PG5 *	I/O	GPIO_Input	USER_DIN_2
91	PG6 *	I/O	GPIO_Input	USER_DIN_1
92	PG7 *	I/O	GPIO_Input	USER_DIN_7
93	PG8 *	I/O	GPIO_Input	USER_DIN_8
94	VSS	Power		
95	VDD33USB	Power		
96	PC6	I/O	TIM3_CH1	COPROCESSOR_TIMER_SYNC_OUTPUT
97	PC7	I/O	DEBUG_TRGIO	
98	PC8 *	I/O	GPIO_Output	CANBUS_POWER_ENABLE
101	PA9	I/O	SPI2_SCK	
103	PA11	I/O	FDCAN1_RX	
104	PA12	I/O	FDCAN1_TX	
105	PA13(JTMS/SWDIO)	I/O	DEBUG_JTMS-SWDIO	
106	VCAP	Power		
107	VSS	Power		
108	VDD	Power		
109	PA14(JTCK/SWCLK)	I/O	DEBUG_JTCK-SWCLK	
110	PA15(JTDI) *	I/O	GPIO_Output	
111	PC10 *	I/O	GPIO_Output	GATE DRIVER POWER ENABLE
112	PC11 *	I/O	GPIO_Input	GATE DRIVER FAULT
113	PC12 *	I/O	GPIO_Input	GATE DRIVER READY
115	PD1 *	I/O	GPIO_Input	VSENSE_ISO_ADC_INTERRUPT
116	PD2 *	I/O	GPIO_Output	SPI2_CS
117	PD3 *	I/O	GPIO_Input	GATE_DRIVE_PWR2_FEEDBACK
118	PD4 *	I/O	GPIO_Input	GATE_DRIVE_PWR1_FEEDBACK
119	PD5 *	I/O	GPIO_Output	GATE DRIVER RESET
120	VSS	Power		

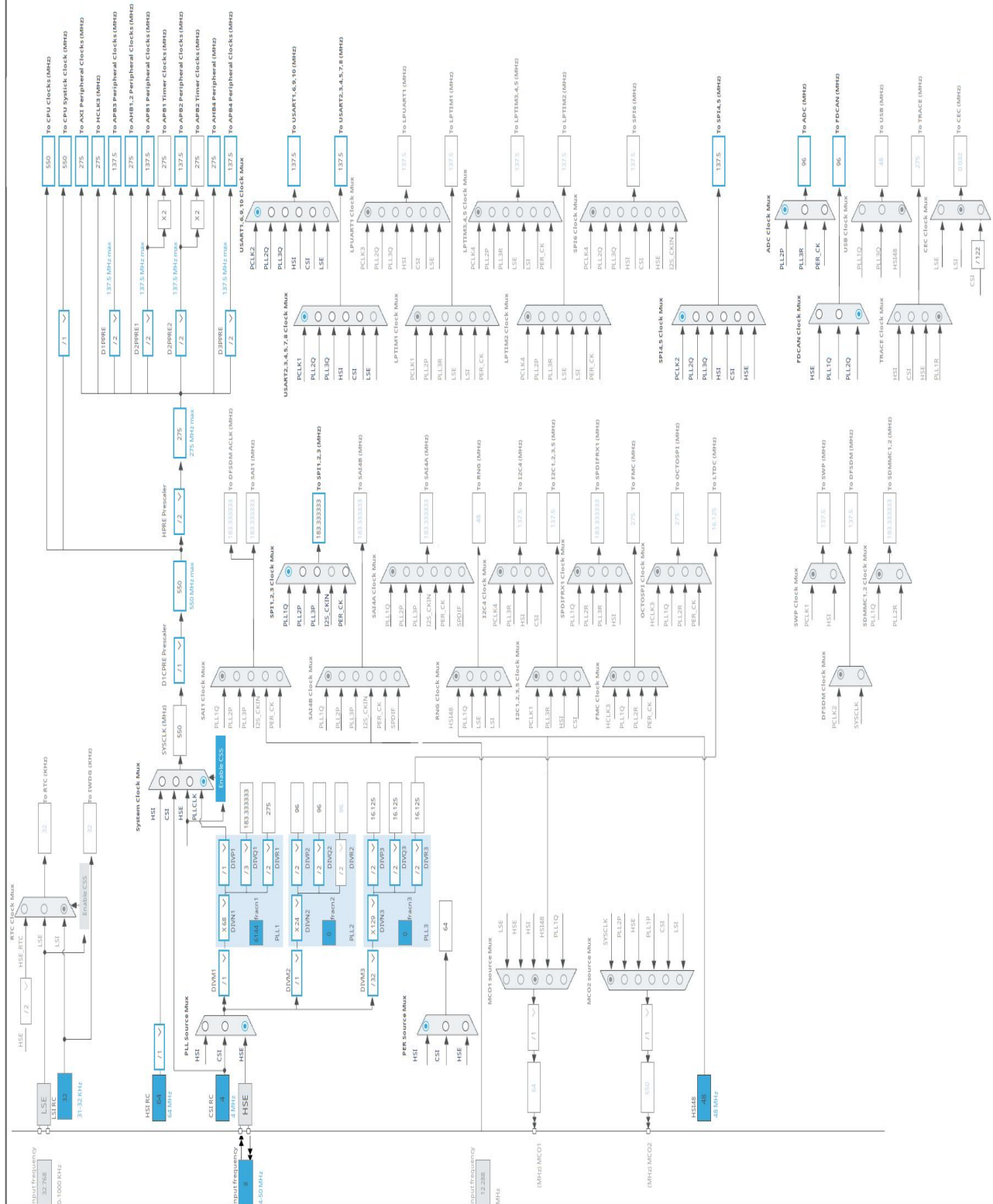
Pin Number LQFP144	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
121	VDD	Power		
126	PG11 *	I/O	GPIO_Output	USER_DOUT_4
127	PG12 *	I/O	GPIO_Output	USER_DOUT_3
128	PG13 *	I/O	GPIO_Output	USER_DOUT_2
129	PG14 *	I/O	GPIO_Output	USER_DOUT_1
130	VSS	Power		
131	VDD	Power		
133	PB3(JTDO/TRACESWO) **	I/O	SPI1_SCK	
134	PB4(NJTRST) **	I/O	SPI1_MISO	
135	PB5 **	I/O	SPI1_MOSI	
136	PB6	I/O	FDCAN2_TX	
138	BOOT0	Boot		
142	PE1 *	I/O	GPIO_Output	DEBUG_ORANGE_LED
143	PDR_ON	Power		
144	VDD	Power		

* The pin is affected with an I/O function

** The pin is affected with a peripheral function but no peripheral mode is activated

4. Clock Tree Configuration

RIF configuration not available for STM32H723ZGTx



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32H7
Line	STM32H723/733
MCU	STM32H723ZGTx
Datasheet	DS13313 Rev1

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Alkaline(9V)
Capacity	625.0 mAh
Self Discharge	0.3 %/month
Nominal Voltage	9.0 V
Max Cont Current	200.0 mA
Max Pulse Current	0.0 mA
Cells in series	1
Cells in parallel	1

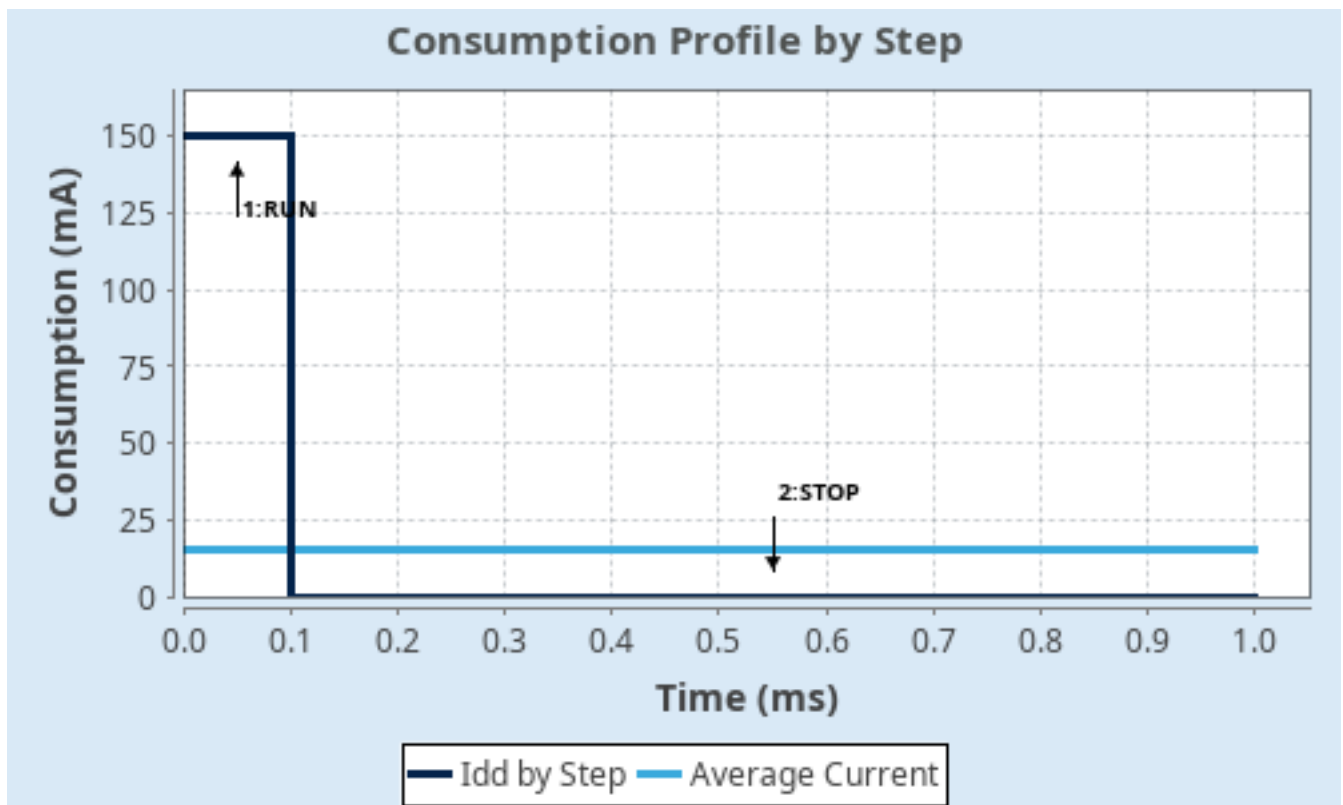
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	VOS0: Scale0/Boost	SVOS5: System-Scale5
D1 Mode	DRUN	DSTANDBY
D2 Mode	DRUN	DSTANDBY
D3 Mode	DRUN	DSTOP
Fetch Type	SRAM1/FlashMode-ON/Cache	NA
CPU Frequency	550 MHz	0 Hz
Clock Configuration	HSE BYP PLL	ALL_CLOCKS_OFF
Clock Source Frequency	8 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	150 mA	94.5 μ A
Duration	0.1 ms	0.9 ms
DMIPS	1177.0	0.0
Ta Max	105.2	124.99
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	15.09 mA
Battery Life	1 day, 17 hours	Average DMIPS	1177.0 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	STM32CubeMX
Project Folder	/home/tliao/Desktop/InverterGen5/STM32CubeMX
Toolchain / IDE	CMake
Firmware Package Name and Version	STM32Cube FW_H7 V1.13.0
Application Structure	Basic
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x1000
Minimum Stack Size	0x2000

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_DMA_Init	DMA
4	MX_USART3_UART_Init	USART3
5	MX_TIM1_Init	TIM1
6	MX_ADC1_Init	ADC1
7	MX_ADC2_Init	ADC2
8	MX_ADC3_Init	ADC3
9	MX_FDCAN1_Init	FDCAN1
10	MX_FDCAN2_Init	FDCAN2
11	MX_SPI4_Init	SPI4

Rank	Function Name	Peripheral Instance Name
12	MX_SPI2_Init	SPI2
13	MX_DAC1_Init	DAC1
14	MX_TIM3_Init	TIM3
15	MX_UART9_Init	UART9

3. Peripherals and Middlewares Configuration

3.1. ADC1

IN2: IN2 Single-ended

IN3: IN3 Single-ended

IN4: IN4 Single-ended

IN5: IN5 Single-ended

mode: IN6

mode: IN7

mode: IN8

mode: IN9

IN16: IN16 Single-ended

mode: IN17

3.1.1. Parameter Settings:

ADCs_Common_Settings:

Mode

Dual regular simultaneous mode only *

DMA Access Mode

DMA access mode disabled

Delay between 2 sampling phases

5,5 Cycles *

ADC_Settings:

Clock Prescaler

Asynchronous clock mode divided by 4 *

Resolution

ADC 16-bit resolution

Scan Conversion Mode

Disabled

Continuous Conversion Mode

Disabled

Discontinuous Conversion Mode

Disabled

End Of Conversion Selection

End of single conversion

Overrun behaviour

Overrun data preserved

Left Bit Shift

No bit shift

Conversion Data Management Mode

Regular Conversion data stored in DR register only

Low Power Auto Wait

Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions

Enable

Enable Regular Oversampling

Enable *

Oversampling Right Shift

No bit shift for oversampling

Oversampling Ratio

16 *

Regular Oversampling Mode

Oversampling Resumed Mode *

Triggered Regular Oversampling

Single trigger for all oversampled conversions

Number Of Conversion

1

External Trigger Conversion Source

Regular Conversion launched by software

External Trigger Conversion Edge	None
<u>Rank</u>	1
Channel	Channel 4 *
Sampling Time	32.5 Cycles *
Offset Number	No offset
Offset Signed Saturation	Disable
ADC_Injected_ConversionMode:	
Enable Injected Conversions	Disable
Analog Watchdog 1:	
Enable Analog WatchDog1 Mode	false
Analog Watchdog 2:	
Enable Analog WatchDog2 Mode	false
Analog Watchdog 3:	
Enable Analog WatchDog3 Mode	false

3.2. ADC2

IN2: IN2 Single-ended

IN3: IN3 Single-ended

mode: IN6

mode: IN7

mode: IN8

mode: IN9

IN10: IN10 Single-ended

mode: IN11

mode: IN14

mode: IN15

3.2.1. Parameter Settings:

ADCs_Common_Settings:

Mode	Dual regular simultaneous mode only *
DMA Access Mode	DMA access mode disabled
Delay between 2 sampling phases	5,5 Cycles *

ADC_Settings:

Clock Prescaler	Asynchronous clock mode divided by 4 *
Resolution	ADC 16-bit resolution
Scan Conversion Mode	Disabled
Continuous Conversion Mode	Disabled

Discontinuous Conversion Mode	Disabled
End Of Conversion Selection	End of single conversion
Overrun behaviour	Overrun data preserved
Left Bit Shift	No bit shift
Conversion Data Management Mode	Regular Conversion data stored in DR register only
Low Power Auto Wait	Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions	Enable
Enable Regular Oversampling	Enable *
Oversampling Right Shift	No bit shift for oversampling
Oversampling Ratio	16 *
Regular Oversampling Mode	Oversampling Resumed Mode *
Triggered Regular Oversampling	Single trigger for all oversampled conversions
Number Of Conversion	1
<u>Rank</u>	1
Channel	Channel 8 *
Sampling Time	32.5 Cycles *
Offset Number	No offset
Offset Signed Saturation	Disable

ADC_Injected_ConversionMode:

Enable Injected Conversions	Disable
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Analog Watchdog 1:

Enable Analog WatchDog1 Mode	false
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Analog Watchdog 2:

Enable Analog WatchDog2 Mode	false
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Analog Watchdog 3:

Enable Analog WatchDog3 Mode	false
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3.3. ADC3

mode: IN0

IN1: IN1 Single-ended

mode: IN7

mode: IN8

mode: IN9

3.3.1. Parameter Settings:

ADC_Settings:

Clock Prescaler	Asynchronous clock mode divided by 1
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Resolution	ADC 12-bit resolution
Scan Conversion Mode	Disabled
Data Alignment	Right alignment
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
DMA Continuous Requests	Disabled
End Of Conversion Selection	End of single conversion
Overrun behaviour	Overrun data preserved
Left Bit Shift	No bit shift
Conversion Data Management Mode	Regular Conversion data stored in DR register only
Low Power Auto Wait	Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions	Enable
Enable Regular Oversampling	Disable
Oversampling Ratio	Oversampling ratio 2x
Number Of Conversion	1
External Trigger Conversion Source	Regular Conversion launched by software
External Trigger Conversion Edge	None
Sampling Mode	Normal
Rank	1
Channel	Channel 1 *
Sampling Time	2.5 Cycles
Offset Number	No offset
Offset Sign	Offset Sign Negative

ADC_Injected_ConversionMode:

Enable Injected Conversions	Disable
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Analog Watchdog 1:

Enable Analog WatchDog1 Mode	false
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Analog Watchdog 2:

Enable Analog WatchDog2 Mode	false
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Analog Watchdog 3:

Enable Analog WatchDog3 Mode	false
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3.4. DAC1

OUT1 connected to: only external pin

OUT2 connected to: only external pin

3.4.1. Parameter Settings:

DAC Out1 Settings:

Mode selected	Normal Mode
Output Buffer	Enable
Trigger	None
User Trimming	Factory trimming

DAC Out2 Settings:

Mode selected	Normal Mode
Output Buffer	Enable
Trigger	None
User Trimming	Factory trimming

3.5. DEBUG

Debug: Serial Wire

mode: External Trigger Bi-directional

3.6. FDCAN1

mode: Activated

3.6.1. Parameter Settings:

Basic Parameters:

Frame Format	Classic mode
Mode	Normal mode
Auto Retransmission	Disable
Transmit Pause	Disable
Protocol Exception	Disable
Nominal Sync Jump Width	1
Data Prescaler	1
Data Sync Jump Width	1
Data Time Seg1	1
Data Time Seg2	1
Message Ram Offset	0
Std Filters Nbr	0
Ext Filters Nbr	0
Rx Fifo0 Elmts Nbr	0
Rx Fifo0 Elmt Size	8 bytes data field
Rx Fifo1 Elmts Nbr	0
Rx Fifo1 Elmt Size	8 bytes data field
Rx Buffers Nbr	0
Rx Buffer Size	8 bytes data field
Tx Events Nbr	0
Tx Buffers Nbr	0

Tx Fifo Queue Elmts Nbr	0
Tx Fifo Queue Mode	FIFO mode
Tx Elmt Size	8 bytes data field

Clock Calibration Unit:

Clock Calibration	Disable
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Bit Timings Parameters:

Nominal Prescaler	8 *
Nominal Time Quantum	83.33333333333333 *
Nominal Time Seg1	17 *
Nominal Time Seg2	6 *
Nominal Time for one Bit	2000 *
Nominal Baud Rate	500000 *

3.7. FDCAN2

mode: Activated

3.7.1. Parameter Settings:

Basic Parameters:

Frame Format	Classic mode
Mode	Normal mode
Auto Retransmission	Disable
Transmit Pause	Disable
Protocol Exception	Disable
Nominal Sync Jump Width	1
Data Prescaler	1
Data Sync Jump Width	1
Data Time Seg1	1
Data Time Seg2	1
Message Ram Offset	0
Std Filters Nbr	0
Ext Filters Nbr	0
Rx Fifo0 Elmts Nbr	0
Rx Fifo0 Elmt Size	8 bytes data field
Rx Fifo1 Elmts Nbr	0
Rx Fifo1 Elmt Size	8 bytes data field
Rx Buffers Nbr	0
Rx Buffer Size	8 bytes data field
Tx Events Nbr	0
Tx Buffers Nbr	0

Tx Fifo Queue Elmts Nbr	16 *
Tx Fifo Queue Mode	FIFO mode
Tx Elmt Size	8 bytes data field

Clock Calibration Unit:

Clock Calibration	Disable
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Bit Timings Parameters:

Nominal Prescaler	8 *
Nominal Time Quantum	83.33333333333333 *
Nominal Time Seg1	17 *
Nominal Time Seg2	6 *
Nominal Time for one Bit	2000 *
Nominal Baud Rate	500000 *

3.8. MEMORYMAP

mode: Activated

3.9. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

3.9.1. Parameter Settings:

Power Parameters:

SupplySource	PWR_LDO_SUPPLY
Power Regulator Voltage Scale	Power Regulator Voltage Scale 0

RCC Parameters:

TIM Prescaler Selection	Disabled
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000
CSI Calibration Value	16
HSI Calibration Value	64

System Parameters:

VDD voltage (V)	3.3
Flash Latency(WS)	3 WS (4 CPU cycle)

PLL range Parameters:

PLL1 input frequency range	Between 8 and 16 MHz
PLL2 input frequency range	Between 8 and 16 MHz
PLL1 clock Output range	Wide VCO range
PLL2 clock Output range	Wide VCO range

3.10. SPI2

Mode: Full-Duplex Master

3.10.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	8 Bits *
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	32 *
Baud Rate	5.729166 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

Advanced Parameters:

CRC Calculation	Disabled
NSSP Mode	Disabled *
NSS Signal Type	Software
Fifo Threshold	Fifo Threshold 01 Data
Tx Crc Initialization Pattern	All Zero Pattern
Rx Crc Initialization Pattern	All Zero Pattern
Nss Polarity	Nss Polarity Low
Master Ss Idleness	00 Cycle
Master Inter Data Idleness	00 Cycle
Master Receiver Auto Susp	Disable
Master Keep Io State	Master Keep Io State Disable
IO Swap	Disabled

3.11. SPI4

Mode: Full-Duplex Master

3.11.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	8 Bits *
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	4 *
Baud Rate	34.375 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge
Advanced Parameters:	
CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Software
Fifo Threshold	Fifo Threshold 01 Data
Tx Crc Initialization Pattern	All Zero Pattern
Rx Crc Initialization Pattern	All Zero Pattern
Nss Polarity	Nss Polarity Low
Master Ss Idleness	00 Cycle
Master Inter Data Idleness	00 Cycle
Master Receiver Auto Susp	Disable
Master Keep Io State	Master Keep Io State Disable
IO Swap	Disabled

3.12. SYS

Timebase Source: SysTick

3.13. TIM1

Channel1: PWM Generation CH1 CH1N

Channel2: PWM Generation CH2 CH2N

Channel3: PWM Generation CH3 CH3N

Activate-Break-Input: Activate Break Input

3.13.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Center Aligned mode1 *
Counter Period (AutoReload Register - 16 bits value)	27500 *
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	Enable *

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Trigger Event Selection TRGO2

Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Enable
BRK Polarity	Low *
BRK Filter (4 bits value)	0
BRK Sources Configuration	
- Digital Input	Enable
Digital Input Polarity	Polarity High
- COMP1	Disable
- COMP2	Disable
- DFSDM	Disable

Break And Dead Time management - BRK2 Configuration:

BRK2 State	Disable
BRK2 Polarity	High
BRK2 Filter (4 bits value)	0
BRK2 Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- DFSDM	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Enable *
Off State Selection for Idle Mode (OSSI)	Enable *
Lock Configuration	Off
Dead Time	194 *

Clear Input:

Clear Input Source	Disable
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PWM Generation Channel 1 and 1N:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CHN Polarity	High
CH Idle State	Reset
CHN Idle State	Reset

PWM Generation Channel 2 and 2N:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable

Fast Mode	Disable
CH Polarity	High
CHN Polarity	High
CH Idle State	Reset
CHN Idle State	Reset

PWM Generation Channel 3 and 3N:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CHN Polarity	High
CH Idle State	Reset
CHN Idle State	Reset

3.14. TIM3

Trigger Source: ITR0

Channel1: PWM Generation CH1

mode: One Pulse Mode

3.14.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable
Slave Mode Controller	Slave mode disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Clear Input:

Clear Input Source	Disable
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PWM Generation Channel 1:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

3.15. UART9

Mode: Asynchronous

3.15.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.16. USART3

Mode: Asynchronous

3.16.1. Parameter Settings:

Basic Parameters:

Baud Rate	460800 *
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC1	PA0	ADC1_INP16	Analog mode	No pull-up and no pull-down	n/a	AIN_TMP_SENSE_3
	PA1	ADC1_INP17	Analog mode	No pull-up and no pull-down	n/a	AIN_TMP_SENSE_2
	PA6	ADC1_INP3	Analog mode	No pull-up and no pull-down	n/a	AIN_PH_V_CURSENS
	PA7	ADC1_INN3	Analog mode	No pull-up and no pull-down	n/a	AIN_PH_V_CURSENS_REF
	PC4	ADC1_INP4	Analog mode	No pull-up and no pull-down	n/a	AIN_PH_U_CURSENS
	PC5	ADC1_INN4	Analog mode	No pull-up and no pull-down	n/a	AIN_PH_U_CURSENS_REF
	PB0	ADC1_INN5	Analog mode	No pull-up and no pull-down	n/a	AIN_PH_W_CURSENS_REF
	PB1	ADC1_INP5	Analog mode	No pull-up and no pull-down	n/a	AIN_PH_W_CURSENS
	PF11	ADC1_INP2	Analog mode	No pull-up and no pull-down	n/a	AIN_DC_LINK_CURSENS
	PF12	ADC1_INN2	Analog mode	No pull-up and no pull-down	n/a	AIN_DC_LINK_CURSENS_REF
ADC2	PC0	ADC2_INP10	Analog mode	No pull-up and no pull-down	n/a	AIN_ENCODER_SIN_HALL_U
	PC1	ADC2_INP11	Analog mode	No pull-up and no pull-down	n/a	AIN_ENCODER_COS_HALL_V
	PA2	ADC2_INP14	Analog mode	No pull-up and no pull-down	n/a	AIN_HALL_W
	PA3	ADC2_INP15	Analog mode	No pull-up and no pull-down	n/a	AIN_THROTTLE_A
	PA6	ADC2_INP3	Analog mode	No pull-up and no pull-down	n/a	AIN_PH_V_CURSENS
	PA7	ADC2_INN3	Analog mode	No pull-up and no pull-down	n/a	AIN_PH_V_CURSENS_REF
	PC5	ADC2_INP8	Analog mode	No pull-up and no pull-down	n/a	AIN_PH_U_CURSENS_REF
	PB0	ADC2_INP9	Analog mode	No pull-up and no pull-down	n/a	AIN_PH_W_CURSENS_REF
	PF13	ADC2_INP2	Analog mode	No pull-up and no pull-down	n/a	AIN_PH_W_CURSENS
	PF14	ADC2_INN2	Analog mode	No pull-up and no pull-down	n/a	AIN_PH_W_CURSENS_REF
ADC3	PF4	ADC3_INP9	Analog mode	No pull-up and no pull-down	n/a	AIN_MOTOR_TMP
	PF6	ADC3_INP8	Analog mode	No pull-up and no pull-down	n/a	AIN_THROTTLE_B
	PF8	ADC3_INP7	Analog mode	No pull-up and no pull-down	n/a	AIN_TMP_SENSE_1
	PC2_C	ADC3_INN1	Analog mode	No pull-up and no pull-down	n/a	AIN_PH_W_CURSENS_REF
	PC3_C	ADC3_INP1	Analog mode	No pull-up and no pull-down	n/a	AIN_PH_W_CURSENS
DAC1	PA4	DAC1_OUT1	Analog mode	No pull-up and no pull-down	n/a	PHASE_CURRENT_COMPARIATOR_N
	PA5	DAC1_OUT2	Analog mode	No pull-up and no pull-down	n/a	DC_LINK_CURRENT_CO

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
						MPARITOR_N
DEBUG	PC7	DEBUG_TRGIO	n/a	n/a	n/a	
	PA13(JTMS/SWDIO)	DEBUG_JTMS-SWDIO	n/a	n/a	n/a	
	PA14(JTCK/SWCLK)	DEBUG_JTCK-SWCLK	n/a	n/a	n/a	
FDCAN1	PA11	FDCAN1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA12	FDCAN1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
FDCAN2	PB12	FDCAN2_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB6	FDCAN2_TX	Alternate Function Push Pull	No pull-up and no pull-down	High *	
RCC	PH0-OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	
	PH1-OSC_OUT	RCC_OSC_OUT	n/a	n/a	n/a	
SPI2	PB14	SPI2_MISO	Alternate Function Push Pull	No pull-up and no pull-down	High *	
	PB15	SPI2_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	High *	
	PA9	SPI2_SCK	Alternate Function Push Pull	No pull-up and no pull-down	High *	
SPI4	PE2	SPI4_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low	FRAM_SPI_SCK
	PE5	SPI4_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Low	FRAM_SPI_MISO
	PE6	SPI4_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Low	FRAM_SPI_MOSI
TIM1	PE8	TIM1_CH1N	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	PH_U_HIGH
	PE9	TIM1_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	PH_U_LOW
	PE10	TIM1_CH2N	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	PH_V_HIGH
	PE11	TIM1_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	PH_V_LOW
	PE12	TIM1_CH3N	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	PH_W_HIGH
	PE13	TIM1_CH3	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	PH_W_LOW
	PE15	TIM1_BKIN	Alternate Function Open Drain	Pull-up *	Low	GATE DRIVER FAULT PWM BREAK
TIM3	PC6	TIM3_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	COPROCESSOR_TIMER_SYNC_OUTPUT
UART9	PG0	UART9_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PG1	UART9_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USART3	PB10	USART3_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB11	USART3_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
Single Mapped Signals	PB3(JTDO/TRACESWO)	SPI1_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Medium *	
	PB4(NJTRST)	SPI1_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Medium *	
	PB5	SPI1_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Medium *	
GPIO	PE3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DEBUG_GREEN_LED
	PC13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	FRAM_WP
	PC14-OSC32_IN	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	FRAM_CS
	PC15-OSC32_OUT	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	FRAM_HOLD
	PF2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	PERIPHERAL_POWER_ENABLE
	PD8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	COPROCESSOR_SYNC
	PD9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	COPROCESSOR_WAKEUP
	PD10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	COPROCESSOR_RESET
	PD11	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	COPROCESSOR_BOOTSEL
	PD15	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	USER_DIN_6
	PG2	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	USER_DIN_5
	PG3	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	USER_DIN_4
	PG4	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	USER_DIN_3
	PG5	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	USER_DIN_2
	PG6	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	USER_DIN_1
	PG7	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	USER_DIN_7
	PG8	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	USER_DIN_8
	PC8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	CANBUS_POWER_ENABLE
	PA15(JTDI)	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Medium *	
	PC10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GATE DRIVER POWER ENABLE
	PC11	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GATE DRIVER FAULT
	PC12	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GATE DRIVER READY
	PD1	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	VSENSE_ISO_ADC_INTERRUPT
	PD2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SPI2_CS
	PD3	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GATE_DRIVE_PWR2_FEEDBACK
	PD4	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GATE_DRIVE_PWR1_FEEDBACK
	PD5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GATE DRIVER RESET

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PG11	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	USER_DOUT_4
	PG12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	USER_DOUT_3
	PG13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	USER_DOUT_2
	PG14	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	USER_DOUT_1
	PE1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DEBUG_ORANGE_LED

4.2. DMA configuration

DMA request	Stream	Direction	Priority
USART3_TX	DMA1_Stream0	Memory To Peripheral	Low

USART3_TX: DMA1_Stream0 DMA request Settings:

Mode: Normal
 Use fifo: Disable
 Peripheral Increment: Disable
 Memory Increment: **Enable ***
 Peripheral Data Width: Byte
 Memory Data Width: Byte

4.3. BDMA configuration

nothing configured in DMA service

4.4. MDMA configuration

nothing configured in DMA service

4.5. NVIC configuration

4.5.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
DMA1 stream0 global interrupt	true	0	0
USART3 global interrupt	true	0	0
PVD/AVD through EXTI Line detection Interrupt	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
ADC1 and ADC2 global interrupts	unused		
FDCAN1 interrupt 0	unused		
FDCAN2 interrupt 0	unused		
FDCAN1 interrupt 1	unused		
FDCAN2 interrupt 1	unused		
TIM1 break interrupt	unused		
TIM1 update interrupt	unused		
TIM1 trigger and commutation interrupts	unused		
TIM1 capture compare interrupt	unused		
TIM3 global interrupt	unused		
SPI2 global interrupt	unused		
TIM6 global interrupt, DAC1_CH1 and DAC1_CH2 underrun error interrupts	unused		
FDCAN calibration unit interrupt	unused		
FPU global interrupt	unused		
SPI4 global interrupt	unused		
HSEM1 global interrupt	unused		
ADC3 global interrupt	unused		
UART9 global interrupt	unused		

4.5.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true
DMA1 stream0 global interrupt	false	true	true
USART3 global interrupt	false	true	true

* User modified value

5. System Views

5.1. Category view

5.1.1. Current

Middleware									
System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing	Trace and Debug	Power and Thermal	Other
BDMA	ADC1 ✓	TIM1 ✓	FDCAN1 ✓				DEBUG ✓		
CORTEX_M7	ADC2 ✓	TIM3 ✓	FDCAN2 ✓						
DMA ✓	ADC3 ✓		SPI2 ✓						
GPIO ⚠	DAC1 ✓		SPI4 ✓						
MDMA			UART9 ✓						
NVIC ✓			USART3 ✓						
RCC ✓									
SYS ✓									

6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32h7-bsdl.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32h7_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32h7-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers_stm32h7_series_product_overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers_stm32h72x-3x_line_product-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h7rs-lines-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/brstm32h7.pdf
Brochures	https://www.st.com/resource/en/brochure/humanoid-robot-reference-guide.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32h7rs.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf

Security Bulletin	https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-protection-statement-for-stmicroelectronics-certified-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4839-level-1-cache-on-stm32f7-series-and-stm32h7-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4891-stm32h72x-stm32h73x-and-singlecore-stm32h74x75x-system-architecture-and-performance-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4990-getting-started-with-sigmadelata-digital-interface-on-applicable-stm32-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5033-stm32cube-mcu-package-examples-for-stm32h7-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5073-receiving-spdif-audio-stream-with-the-stm32f4f7h7-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5612-esd-protection-

of-stm32-mcus-and-mpus-stmicroelectronics.pdf

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Application Notes https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf

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Application Notes https://www.st.com/resource/en/application_note/an5225-introduction-to-usb-typec-power-delivery-for-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5342--how-to-use-error-correction-code-ecc-management-for-internal-memories-protection-on-stm32-mcus-stmicroelectronics.pdf

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Application Notes https://www.st.com/resource/en/application_note/an3126-how-to-use-dac-for-audio-and-waveform-generation-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

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Application Notes https://www.st.com/resource/en/application_note/an3156-how-to-use-usb-ldfu-protocol-in-bootloader-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4286-how-to-use-spi-protocol-in-bootloader-on-stm32-mcus-stmicroelectronics.pdf

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